

IL - AE - TPF Interview - Prepare for Technical Interview Round (TPF) (Send-to-Candidate) - v2.1

Overview

This document is for candidates who have passed the initial screening and are preparing for the Analytics Engineering - Technical Progression Framework (AE - TPF) interview. It explains what you are expected to do, how the session runs, and how you will be assessed, so that you can prepare effectively.

Please note:

- This document should be sent to you at least **7 days before** your scheduled Technical (TPF) interview.
- It is designed to be self-contained. Due to time constraints, we **will not answer additional questions** about this document before the interview.

What you need to do (in short)

This is an evidence-based interview, and you choose and bring the work you will be assessed on. We do not assign a specific project or task. That said, the work is not entirely free choice: it must let you demonstrate the **05 core skills** on the **required technology stack** (see "Core Evaluation Areas & Technology Stack"), with live evidence, within the time available (see "Structure & Timing"). Choosing or preparing work that covers these areas is the first step.

A few things that will help you prepare:

- Bring work you can demonstrate and explain live. It does not have to be newly built or solely your own. See "**Proof of Technical Knowledge**" for what qualifies.
- We focus on the evidence and your reasoning rather than the size of the project (see "**Assessment Principles**"). A focused solution you understand well and can extend on the spot tends to score higher than a large one that you are unable to navigate or explain.
- You will not have time to cover everything in equal detail, so it helps to plan where to focus: prioritize work that spans several areas, go deeper where you are strongest, and let the interviewer know when you are ready to move on. Breadth and depth matter equally (see "Project Scope" and "Time Management & Maximizing Your Score").

What this interview looks like (Show → Why → What-if)

This is an **evidence-based** interview. For each topic (core evaluation area, see "Core Evaluation Areas & Technology Stack") you will typically be asked to:

- **Show:** where/how you implemented it (code/models/tests/BI/PRs/diagrams, or other credible evidence).
- **Why:** why you made those decisions (trade-offs, alternatives, constraints)
- **What-if:** adapt to a small change or new requirement (live extension/modification when reasonable)

Points are awarded based on the quality of evidence and how well you can explain, extend, or redesign your work (see "Assessment Principles"). If you haven't implemented a topic in advance, you may implement a small example live (time-boxed live coding, preferably using your own environment).

Proof of Technical Knowledge

To fairly assess your technical skills, the interview requires a practical demonstration of your knowledge (via Show → Why → What-if, above). The most straightforward way to achieve this is by preparing a project or piece(s) of work that you can showcase live during the interview.

- This does **not** have to be a brand-new project created specifically for this interview.
- It can be one project, multiple projects, or individual components from work you've done previously - as long as each piece can serve as proof for the core areas (see "Project Scope").
- What matters most is that you can **demonstrate** and **explain** your work live, including code walkthroughs and answering related questions.

✖ Ideally, the work should be entirely your own and reflect your design and implementation decisions, as this best demonstrates your technical judgment and depth of understanding. We do accept team-based or assigned work, but such examples may limit how much you can showcase your end-to-end design ownership.

✔ This interview has specific requirements (see "Project Scope" and "Core Evaluation Areas & Technology Stack"). It is worth reviewing your chosen work against them in advance, as described under "Project Scope".

Project Scope

Select or assemble an end-to-end data solution focused on solving a business problem with a single meaningful, quantifiable metric (e.g., Profit, Cost, Revenue, LTV, ROAS...).

- The scope and complexity are your choice, but your solution must be fully functional and ready for live demonstration within the interview duration (see "Structure & Timing").
- Balance your approach between:

- **Depth** (detailed and complex logic)
- **Breadth** (covering all core evaluation areas)
- Before the interview, it helps to map your chosen work against the five Core Evaluation Areas and the Required Technology Stack below. Where it falls short, you can decide whether to build a small addition or to focus your time on areas where you are stronger.

Core Evaluation Areas & Technology Stack

Your solution will be assessed on your ability to understand, implement, and design across the five core areas below, demonstrated on the `required technology stack`.

Required Technology Stack

- **Transformation:** dbt Core or dbt Cloud.
- **Data Warehouse:** A cloud data warehouse (e.g., Snowflake, BigQuery, Databricks).
- **BI/Analytics:** A BI tool (e.g., Omni, Sigma, Power BI, Looker) connected directly to your warehouse models.

Core evaluation areas

We will evaluate your project based on these 05 core skills. Depending on your project/proof, the points listed in each skill below are included, but not limited to, what we may explore during the session:

1. Data Modeling & Architecture

- Implement a modeling approach (e.g., Dimensional, Data Vault, 3NF) with clear layered structures (e.g., Medallion Architecture).

2. Data Transformation

- Build dbt transformations, including at least one incremental model.
- Incorporate ELT/ETL best practice with dbt (deduplication, dbt macros, other advanced dbt techniques...)

3. Data Governance (Quality & Security)

- Implement live data quality checks via at least one dbt test(s) or warehouse-native features.
- Apply appropriate handling of sensitive data (masking, hashing, or access control).

4. Data Insight

- Present final business metric(s) through your chosen BI tool's dashboard or semantic layer.
- Explain **how you calculate these metrics**, including the SQL logic used (joins, aggregations, filters applied, etc.) and the business impact of the metrics.

5. DataOps

- Demonstrate how you manage code quality and deployment via your git branching strategy, CI/CD pipeline, release process, and other practices.

i During the interview, you will be expected to answer deep technical questions and may be asked to extend or modify your solution live to demonstrate your understanding.

Practical checklist

- A repo/project/artifact you can navigate quickly.
- A **working** dbt project you can run (models + tests). You will be asked to execute **dbt** **commands** to prove that your project is working as intended, as mentioned in Data Transformation.
- Access to your data warehouse to demonstrate that your data ingestion and transformation processes are working correctly.
- Access to your BI tool & related dashboard(s) so you can demo your Data Insight requirements.
- A prepared demo plan with clear time management (what you will show first, where the evidence lives, and how long you should spend on each section). See “Time Management and Maximizing Your Score”.
- Evidence for Data Modeling & Architecture (ERDs, documentation/ walkthrough... explaining data modelling concept of choice)
- Evidence for DataOps (e.g., CI/CD config, run logs, screenshots, or a walkthrough of your branching/release flow).

Confidentiality & Safety

- Do **not** share passwords, secrets, tokens, or private keys on screen during the demo.
- Recommended to use **sanitized/redacted** configs and **non-sensitive** datasets.
- If something can't be shown due to confidentiality, say so and **provide alternative evidence** (diagram, pseudo-config, redacted screenshot, or explanation of the control and how it is enforced).

Interview Guidelines

Structure & Timing (Approximate)

- **05 minutes:** Overview and introduction
- **75 minutes (15 minutes per topic x 5 topics):** Walkthrough of your solution (using your project as proof), including live extensions or modifications as needed

- **10 minutes:** Q&A and further evaluation
- **Total session:** Will not exceed 1.5 hours

Interview Flow & Interruptions

- Interviewers may interrupt to manage time or move forward once they have enough information. These interruptions **do not negatively affect your evaluation**.
- Interviewers will take notes during the session as standard practice.

Time Management & Maximizing Your Score

 It is recommended to follow this order of topics during your presentation:

0. Overview
1. Data Modeling & Architecture
2. Data Transformation
3. Data Governance (Quality & Security)
4. Data Insight
5. DataOps

This helps interviewers follow along easily. You're welcome to use a different format, just outline your flow at the start so the interviewer knows what to expect.

- You have the flexibility to **control the flow**. If you want to spend more time on a particular area to highlight your strengths, or skip a question to move on to another topic, feel free to let the interviewer(s) know.
- Aim to keep answers concise and grounded in **concrete evidence** from your work.
- As a rule of thumb, expect to spend around **5 minutes per question or sub-topic**.
- Each core evaluation area typically includes **up to 3 sub-topics**, depending on your project/proof and the interviewer(s).
- Each question will be rated on a **0-3 point scale**, with total scores averaged across all core evaluation areas.

AI Usage Policy

AI tools are **strictly prohibited DURING the live interview**, including screen sharing, live coding, Q&A, and any other interview activity. Any detected use of AI during the interview will result in **immediate disqualification**.

Exception: if live coding is required, AI-assisted IDE autocomplete and AI inline syntax suggestions are allowed.

Prohibited AI usage includes (but is not limited to):

- **ChatGPT, Claude, Gemini, or similar tools**, including any generative AI or (large/small) language model-based assistants such as Copilot, Cursor, Claude Code, Windsurf, or equivalent AI-powered chat, coding, search, or reasoning tools.
- **AI-generated code or explanations used during the interview**, including code, queries, solutions, architectural designs, written explanations, or verbal answers that are generated, suggested, or refined by AI systems **in real time**.
- **AI-assisted IDE features or live copy-pasting of AI-generated content**, including IDE-integrated AI autocomplete, inline suggestions, refactoring tools, chat panels, browser extensions, or copying/pasting content produced by AI tools during the interview session.

AI tools are permitted for preparation *BEFORE* the interview, such as studying, practicing, writing docs, and building your project or repository. In short, you can build your project with AI assistance, but you must present the project in the interview without it. All work shown during the interview must reflect your own understanding and ability.

Assessment Principles

- Evaluation focuses on your ability to understand, implement, justify, and extend your solution live.
- Both vertical depth (seniority signal) and horizontal breadth (coverage of core areas) matter equally.
- Success depends on balancing complexity, functionality, and clear communication within the available time.